



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 9 • STANDARD 6.AT.11

Explore Relationships Between Two Variables

Lesson 9-1 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can recognize when two quantities change together, and identify which one is the independent variable and which is the dependent variable.

Language Objective: I can explain which quantity depends on the other using the words independent variable, dependent variable, and depends on.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

independent variable

dependent variable

depends on

quantity

relationship



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

— The quantity that is subject to choice. It does not depend on the other quantity.

2

— The quantity that changes in response to the independent variable.

3

— One quantity depends on another when changing the first causes the second to change.

4

— Something that can be counted or measured.

5

— How two quantities change together.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Which two quantities in this shop are related, so that one depends on the other — and which quantity does NOT depend on the number of pizzas at all?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- ☐ **independent variable** — The quantity that is subject to choice. It does not depend on the other quantity.
variable independiente — La cantidad que se elige. No depende de la otra cantidad.

- ☐ **dependent variable** — The quantity that changes in response to the independent variable.
variable dependiente — La cantidad que cambia en respuesta a la variable independiente.

- ☐ **depends on** — One quantity depends on another when changing the first causes the second to change.
depende de — Una cantidad depende de otra cuando cambiar la primera hace cambiar la segunda.

- ☐ **quantity** — Something that can be counted or measured.
cantidad — Algo que se puede contar o medir.

- ☐ **relationship** — How two quantities change together.
relación — Cómo dos cantidades cambian juntas.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

In this scenario, the ____ is the independent variable. The ____ and the ____ both depend on it. The number of ____ stays the same no matter how many pizzas are made.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Akela chose where to place each tray, which set the hours of sunlight; the plant height was measured afterward, in response to the sunlight.

Evidence: A strong answer names sunlight as chosen (independent) and height as responding (dependent), and can say the sentence 'height depends on sunlight.' A weak answer just says 'height is bigger' without naming which quantity was chosen.

Reasoning: This evidence connects the definition of relationship to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- In this scenario, the ____ is the independent variable. The ____ and the ____ both depend on it. The number of ____ stays the same no matter how many pizzas are made.
- My answer is ____.

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- My claim is ____.
Mi afirmación es ____.
- I know because ____.
Lo sé porque ____.
- So ____.
Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** independent variable
2. **Notes 2:** dependent variable
3. **Notes 3:** depends on
4. **Notes 4:** quantity
5. **Notes 5:** relationship
6. **Watch for this mistake:** *Naming the variables backward — calling the responding quantity 'independent' because it is the one the question asks about.*
7. **Try It 1:** A. A student's height and their house number — *A house number is assigned by an address, not by how tall someone is — the two are unrelated.*
8. **Try It 2:** A. Pages printed depends on minutes running — *You choose how long to let it run; the number of pages is the result of that choice.*